

Ten-eleven translocation proteins and their role beyond DNA demethylation - what we can learn from the fly.

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Ten-eleven Translocation (TET) proteins have emerged as a family of epigenetic regulators that are important during development and have been implicated in various types of cancers. TET is a highly conserved protein that has orthologues in almost all multicellular organisms. Here, we review recent literature on the novel substrate specificity of this family of DNA 5-methylcytosine demethylases on DNA 6-methyladenine and RNA 5-methylcytosine that were first identified in the invertebrate model *Drosophila*. We focus on the biological role of these novel epigenetic marks in the fruit fly and mammals and highlight TET proteins' critical function during development specifically in brain development.

[Roles of ten eleven translocation proteins family and 5-hydroxymethylcytosine in epigenetic regulation of stem cells and regenerative medicine].

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The methylation of cytosine is one of the most fundamental epigenetic modifications in mammalian genomes, and is involved in multiple crucial processes including gene expression, cell differentiation, embryo development and oncogenesis. In the past, DNA methylation was thought to be an irreversible process, which could only be diluted passively through DNA replication. It is now becoming increasingly obvious that DNA demethylation can be an active process and plays a crucial role in biological processes. Ten eleven translocation (TET) proteins are the key factors modulating DNA demethylation. This family contains three members: TET1, TET2 and TET3. Although three TET proteins have relatively conserved catalytic domains, their roles in organisms are not repeated, and their expression has significant cell/organ specificity. TET1 is mainly expressed in embryonic stem cells, TET2 is mainly expressed in hematopoietic system, and TET3 is widely expressed in cerebellum, cortex and hippocampus. This family catalyzes 5-methylcytosine to 5-hydroxymethylcytosine and other oxidative products, reactivates silenced-gene expression, in turn maintains stem cell pluripotency and regulates lineage specification. With the development of tissue engineering, organ transplantation, autologous tissue transplantation and artificial prosthesis have been widely used in clinical treatment, but these technologies have limitations. Regenerative medicine, which uses stem cells and stem cell related factors for treatment, may provide alternative therapeutic strategies for multiple diseases. Among all kinds of human stem cells, adipose-derived stem cells (ADSCs) are the most prospective stem cell lineage since they have no ethical issues and can be easily obtained with large quantities. To date, ADSCs have been shown to have strong proliferation capacity, secrete numerous soluble factors and have multipotent differentiation ability. However, the underlying mechanism of the proliferation, secretion, acquired pluripotency, and lineage specific differentiation of ADSCs are still largely unknown. Some studies have explored the role of epigenetic regulation and TET protein in embryonic stem cells, but little is known about its role in ADSCs. By studying the roles of TET proteins and 5-hydroxymethylcytosine in ADSCs, we could provide new theoretical foundation for the clinical application of ADSCs and the stem cell-based therapy. In the future, combined with bioprinting technology, ADSCs may be used in tissue and organ regeneration, plastic surgery reconstruction and other broader fields.

The TET2 interactors and their links to hematological malignancies.

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Ten-eleven translocation (TET) family proteins are dioxygenases that oxidize 5-methylcytosine to 5-hydroxymethylcytosine, 5-formylcytosine, and 5-carboxylcytosine in DNA, early steps of active DNA demethylation. TET2, the second member of TET protein family, is frequently mutated in patients with hematological malignancies, leading to aberrant DNA methylation profiling and decreased 5hmC levels. Located in the nucleus and acting as a DNA-modifying enzyme, TET2 is thought to exert its function via TET2-containing protein complexes. Identifying the interactome network of TET2 likely holds the key to uncover the mechanisms by which TET2 exerts its function in cells. Here, we review recent literature on TET2 interactors and discuss their possible roles in TET2 loss-mediated dysregulation of hematopoiesis and pathogenesis of hematological malignancies.

DNA methylation: TET proteins-guardians of CpG islands?

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DNA methylation is involved in key cellular processes, including X-chromosome inactivation, imprinting and transcriptional silencing of specific genes and repetitive elements. DNA methylation patterns are frequently perturbed in human diseases such as imprinting disorders and cancer. The recent discovery that the three members of the TET protein family can convert 5-methylcytosine (5mC) into 5-hydroxymethylcytosine (5hmC) has provided a potential mechanism leading to DNA demethylation. Moreover, the demonstration that TET2 is frequently mutated in haematopoietic tumours suggests that the TET proteins are important regulators of cellular identity. Here, we review the current knowledge regarding the function of the TET proteins, and discuss various mechanisms by which they contribute to transcriptional control. We propose that the TET proteins have an important role in regulating DNA methylation fidelity, and that their inactivation contributes to the DNA hypermethylation phenotype often observed in cancer.

TET family proteins: new players in gliomas.

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DNA methylation at the 5-position of cytosine (5mC) in the mammalian genome has emerged as a pivotal epigenetic event that plays important roles in development, aging and disease. The three members of the TET protein family, which convert 5mC to 5-hydroxymethylcytosine, has provided a potential mechanism resulting in DNA demethylation and maintaining cellular identity. Recent studies have shown that epigenetic modifications play a key role in the regulation of the molecular pathogenesis of gliomas. In this review we focus on demonstrating the TET proteins in DNA demethylation and transcriptional regulation of different target genes. In addition, we address the role of TET proteins in gliomas. This review will provide valuable insights into the potential targets of gliomas, and may open the possibility of novel therapeutic approaches to this fatal disease.

Identification of a missense variant in SPDL1 associated with idiopathic pulmonary fibrosis.

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Idiopathic pulmonary fibrosis (IPF) is a fatal disorder characterised by progressive, destructive lung scarring. Despite substantial progress, the genetic determinants of this disease remain incompletely defined. Using whole genome and whole exome sequencing data from 752 individuals with sporadic IPF and 119,055 UK Biobank controls, we performed a variant-level exome-wide association study (ExWAS) and gene-level collapsing analyses. Our variant-level analysis revealed a novel association between a rare missense variant in SPDL1 and IPF (NM_017785.5:g.169588475 G > A p.Arg20Gln; $p = 2.4 \times 10^{-7}$, odds ratio = 2.87, 95% confidence interval: 2.03-4.07). This signal was independently replicated in the FinnGen cohort, which contains 1028 cases and 196,986 controls (combined $p = 2.2 \times 10^{-20}$), firmly associating this variant as an IPF risk allele. SPDL1 encodes Spindly, a protein involved in mitotic checkpoint signalling during cell division that has not been previously described in fibrosis. To the best of our knowledge, these results highlight a novel mechanism underlying IPF, providing the potential for new therapeutic discoveries in a disease of great unmet need.

Defining "mutation" and "polymorphism" in the era of personal genomics.

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The growing advances in DNA sequencing tools have made analyzing the human genome cheaper and faster. While such analyses are intended to identify complex variants, related to disease susceptibility and efficacy of drug responses, they have blurred the definitions of mutation and polymorphism. In the era of personal genomics, it is critical to establish clear guidelines regarding the use of a reference genome. Nowadays DNA variants are called as differences in comparison to a reference. In a sequencing project Single Nucleotide Polymorphisms (SNPs) and DNA mutations are defined as DNA variants detectable in $>1\%$ or $<1\%$ of the population, respectively. The alternative use of the two terms mutation or polymorphism for the same event (a difference as compared with a reference) can lead to problems of classification. These problems can impact the accuracy of the interpretation and the functional relationship between a disease state and a genomic sequence. We propose to solve this nomenclature dilemma by defining mutations as DNA variants obtained in a paired sequencing project including the germline DNA of the same individual as a reference. Moreover, the term mutation should be accompanied by a qualifying prefix indicating whether the mutation occurs only in somatic cells (somatic mutation) or also in the germline (germline mutation). We believe this distinction in definition will help avoid confusion among researchers and support the practice of sequencing the germline and somatic tissues in parallel to classify the DNA variants thus defined as mutations.

PCR-Based Detection Methods for Single-Nucleotide Polymorphism or Mutation: Real-Time PCR and Its Substantial Contribution Toward Technological Refinement.

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Single-nucleotide polymorphisms (SNPs) and single-nucleotide mutations result from the substitution of only a single base. The SNP or mutation can be relevant to disease susceptibility, pathogenesis of

disease, and efficacy of specific drugs. It is important to detect SNPs or mutations clinically. Methods to distinguish/detect SNPs or mutations should be highly specific and sensitive. In this regard, polymerase chain reaction (PCR) has provided the necessary analytical performance for many molecular analyses. PCR-based methods for SNP/mutation detection are broadly categorized into two types-(1) polymorphic or mutant allele-directed specific analysis using primers matched with substituted nucleotide or using oligonucleotides to block or clamp the nontargeted template, and (2) melting curve analysis, which is combined with the real-time PCR techniques using hydrolysis probes, hybridization probes, or double-stranded DNA-binding fluorescent dyes. Innovative and novel approaches as well as technical improvements have made SNP- or mutation-detection methods increasingly more sophisticated. These advances include DNA/RNA preparation and subsequent amplification steps, and miniaturization of PCR instruments such that testing may be performed with relative ease in clinical laboratories or as a point-of-care test in clinical settings.

Detection of single-base mutation in RNA using T4 RNA ligase-based nick-joining or DNAzyme-based nick-generation.

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A single nucleotide polymorphism (SNP) is a common genetic variation when a single nucleotide differs between members of a species or paired chromosome. Due to its association with disease susceptibility and drug resistance, SNP detection is of great value in studying the variation in drug responses. Here we present two quantitative SNP detection methods for a single-base mismatch in RNA, based on nick-joining and nick-generating activities of T4 RNA ligase and DNAzyme, respectively. T4 RNA ligase successfully discriminated a one-base mismatch in the ligation junction, and the designed DNAzyme cleaved RNA by discerning a single-base mismatch in the cleaving site.

Single Nucleotide Polymorphism in SMAD7 and CHI3L1 and Colorectal Cancer Risk.

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Colorectal cancer (CRC) is one of the leading cancers throughout the world. It represents the third most common cancer and the fourth in mortality. Most of CRC are sporadic, arise with no known high-penetrant genetic variation and with no previous family history. The etiology of sporadic CRC is considered to be multifactorial and arises from the interaction of genetic variants of low-penetrant genes and environmental risk factors. The most common well-studied genetic variation is single nucleotide polymorphisms (SNPs). SNP arises as a point mutation. If the frequency of the sequence variation reaches 1% or more in the population, it is referred to as polymorphism, but if it is lower than 1%, the allele is typically considered as a mutation. Lots of SNPs have been associated with CRC development and progression, for example, genes of TGF- β 1 and CHI3L1 pathways. TGF- β 1 is a pleiotropic cytokine with a dual role in cancer development and progression. TGF- β 1 mediates its actions through canonical and noncanonical pathways. The most important negative regulatory protein for TGF- β 1 activity is termed SMAD7. The production of TGF- β can be controlled by another protein called YKL-40. YKL-40 is a glycoprotein with an important role in cancer initiation and metastasis. YKL-40 is encoded by the CHI3L1 gene. The aim of the present review is to give a brief introduction of CRC, SNP, and examples of some SNPs that have been documented to be associated with CRC. We also discuss two important signaling

pathways TGF- β 1 and CHI3L1 that influence the incidence and progression of CRC.
